VR Pub Quiz Project: Technical Plan

KEY TECHNOLOGIES

Hardware
- **VR Headset**: Meta Quest 3 (or equivalent)
- **Controllers**: Meta Quest 3 controllers (or equivalent)
- **PC Requirements**: For development, a computer with specifications supporting Unity VR development

Software
- **Unity Engine**: Core platform for building the VR experience.
- **XR Interaction Toolkit**: For handling VR interactions such as hand gestures, ray interactions, and haptics.
- **Blender/3D Modeling Tools**: For designing custom 3D assets like Monica's apartment and quiz panels.
- **Photon Unity Networking (PUN)**: For multiplayer functionality, including real-time communication and team-based play.

Plugins
- **SteamVR**: To ensure compatibility with a wide range of VR devices.
- **AudioToolkit**: For spatial audio and sound effects.

KEY INTERACTION TECHNIQUES IMPLEMENTATION

Controller-Based Menu Navigation
- **Unity Components**:
    - XRRayInteractor for ray-based selection.
    - Canvas and GraphicRaycaster for interactive menus.
- **Implementation**: Users interact with holographic menus using controller rays, with triggers to select quiz themes or answers.

Touch-Based Answer Selection
- **Unity Components**:
    - XR Interaction Manager and XR Grab Interactor.
- **Implementation**: Players use controllers or gestures to touch holographic buttons on the quiz panel. Haptic feedback is triggered upon selection.

Scene Transitions
- **Unity Techniques**:
    - SceneManager.LoadScene for dynamic scene transitions.
- **Implementation**: Player-selected themes trigger loading of corresponding environments (e.g., Monica's apartment). Transition effects such as fades or zooms maintain immersion.

Real-Time Communication
- **Unity Plugins**:
    - Photon Voice for real-time voice communication.
- **Implementation**: Enables team-based collaboration with voice and chat features.

Haptic Feedback
- **Unity Components**:
    - XRController.SendHapticImpulse.

- **Implementation**: Vibrations occur when users select answers or interact with the environment.

ASSETS AND 3D SCENE DESIGN

3D Models

- **Custom Models**:
  - Monica's apartment and Central Perk cafe: Refer to film set images to design, might have to use Blender to custom make iconic furniture in the scene, or else, rely on Unity Asset store to find closest match.
  - Pub environment: Might need dynamic props like beer mugs that players can pick up.
- **Pre-Built Assets**:
  - Furniture, lighting, and decor sourced from Unity Asset Store where appropriate.

Holographic Quiz Panels

- Designed in Unity Canvas with custom animations for glowing effects.

Scene Layouts

- **Menu for theme selection**: Simple panel design for easy navigation, set in a pub environment.
- **Monica's Apartment**:
  - Quiz panel near the coffee table.
  - Other player avatars around the room.
  - Try to mimic the original layout of the living room in the show as much as possible.
- **Central Perk**:
  - Trivia board behind the counter.
  - Seating areas on sofas for players.

Sound Assets:

- **Background pub sounds, quiz game sounds, button click SFX (Unit Asset Store)**
  - Casual Game Sound FX, Bar crowd ambience pack, FRIENDS theme song

EVALUATION PLAN

Feature Testing

- **Define Testable Features**:
  - Menu navigation, Answer Selection, Scene Transitions, Multiplayer Features, Teleportation
- **Performance Testing:** Test frame rates and responsiveness to ensure smooth gameplay

Usability Testing

- **Participants**: 10-15 users, including FRIENDS fans and general quiz enthusiasts.
- **Metrics**:
  - Ease of interaction with quiz panels.
  - Immersion in themed environments.
  - Quality of multiplayer features (e.g., communication).
  - Simulator sickness

Feedback Collection

- **Methods**:
  - Surveys and interviews post-testing.
    - Ask players how much does the game meet their expectations of a pub quiz.
    - Do they prefer to play the solo mode or social mode with other players.
  - Observational notes during play sessions.

Iterative Improvements

- Address feedback to refine user experience (e.g., interaction speed, visual clarity).